Solar Energy Materials and Solar Cells

Oxidation-Budgeted Design of Lead-Free RbGeI3 Solar Cells: Back-Surface-Field Enhancement, Realistic Loss Budgeting and SHAP Sensitivity From SCAPS-1D Optimization to Manufacturing Tolerances, Benchmarked Against CsGeI3

--Manuscript Draft--

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Abstract:	The rapid rise of lead-halide perovskite solar cells has been tempered by lead toxicity, motivating a search for lead-free absorbers with comparable optoelec- tronics. Rubidium germanium iodide (RbGeI3) offers a direct, tunable bandgap, strong absorption, and improved environmental compatibility. Here we report a SCAPS-1D optimization of an RbGeI3 cell organized around its central liabil- ity, Ge2+ oxidation. After screening eight ETL–HTL combinations, the FTO/ TiO2/RbGeI3/CuI/Au architecture was carried forward on stability and cost grounds (a D2/NiO cross-check through the same pipeline reaches 28.16%, so this stack is not the efficiency optimum). Eleven parameters were varied se- quentially on one frozen deck; the consolidated device reaches 27.06% PCE. A Cu2O-like back-surface field raises the ideal point to 28.01% and recovers about one point on the identical full-realism stack (20.78% → 21.76%). Mapping VGe vacancy thermodynamics onto the defect ladder converts air exposure into a fab spec (Ef ≥ 0.46 eV holds the headline at 300 K equilibrium, 0.57–0.65 eV for anneal freeze-in; ambient falls below 20.76%), with a 15.53% current-state rung marking what first fabrication must beat. The solver pipeline is calibration-checked against experimental β-CsPbI3 [55], and SHAP analysis recovers bulk defects, bandgap and electron mobility as the dominant levers.
Opposed Reviewers:	

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Cover letter — submission to Solar Energy Materials and Solar Cells

Dear Editor,

We submit our revised manuscript "Oxidation-Budgeted Design of Lead-Free RbGeI3 Solar Cells: Back-Surface-Field Enhancement, Realistic Loss Budgeting
and SHAP Sensitivity From SCAPS-1D Optimization to Manufacturing Tolerances,
Benchmarked Against CsGeI3" for reconsideration as a Regular Paper in Solar Energy.

Using SCAPS-1D on one frozen deck, we optimize a lead-free FTO/TiO2/RbGeI3/CuI/Au cell to 27.06% PCE (SQ-checked) and go beyond standard parameter sweeps: a Cu2O-like p+ back-surface field raising the ideal point to 28.01% (+0.98 pp on the identical full-realism stack, 20.78% to 21.76%, via rear-interface gap widening isolated by controlled trap attribution with a p+ control), a D2/NiO cross-check through the same
pipeline (28.16%, so the reported stack trades efficiency for stability), a single stacked loss waterfall bounding the experimentally plausible window
(optics, bulk/interface defects, radiative/Auger bounds, parasitics), defect-energy/capture-asymmetry mapping of Ge-related degradation, background-doping and mobility viability windows recomputed on the stacked point, series/shunt manufacturing tolerances, temperature diagnostics with a disclosed Ea extrapolation artifact (1.48 eV), a lifetime baseline (1000 ns, 8.6 um) for experimental comparison, a held-out XGBoost/SHAP sensitivity analysis recovering bulk defects, bandgap and electron mobility as the top levers, and a same-stack CsGeI3 gap control (fixed-chi protocol, not A-site proof). The work gives experimentalists quantitative targets (BSF NA ~1e19 cm-3, bulk Nt <= 1e15 cm-3, mobility >= 15 cm2/Vs, Rs <= 4 ohm.cm2) rather than an idealized efficiency number.

Crucially, this revision answers the experimental-grounding concern head-on.
No RbGeI3 device has been fabricated to date, so we calibration-check the solver pipeline itself: our SCAPS-1D rebuild of the experimental beta-CsPbI3
devices of Wang et al. (Science 2019) reproduces both the control (15.18 vs 15.10%) and the CHI-passivated champion (18.49 vs 18.40%) within 2% by tuning defect densities (plus one resistive correction), presented as calibration rather than validation. We further place our realistic window against a frozen pre-2021 snapshot of 41,316 reported devices from the Perovskite Database Project (NOMAD) purely as era-bounded
context, with no superiority claim over the present state of the art. Finally, we convert the central liability of this absorber --- Ge2+ oxidation --- into a quantitative fab spec: mapping V_Ge vacancy thermodynamics onto our defect ladder shows E^f >= 0.46 eV (300 K equilibrium; 0.57-0.65 eV for anneal freeze-in) holds 27.06% while ambient

processing falls below 20.76%, with passivation chemistries (PCCMAI, Lewis
bases) acting as Nt reduction in waterfall language.

The work is original, not under consideration elsewhere, and all authors approve submission. A prior version was rejected; this revision rebuilds the contribution around the oxidation budget, BSF variant with measured desensitization mechanism, single stacked loss budget, lifetime baseline, held-out ML surrogate, and experimental anchoring. We suggest reviewers with expertise in lead-free perovskite simulation and SCAPS-1D device modelling.

Yours sincerely,
Hussain Touhid Siddiquee (second and corresponding author)
touhidsiddiqueeraj@gmail.com
on behalf of Md. Abdul Malek Fahim (first author;
mdabdulmalekfahim@gmail.com)
Department of Electrical and Electronic Engineering,
Leading University, Sylhet, Bangladesh

Manuscript word count: ~4,060 words excluding references (body prose ~2,880; tables/figure captions ~1,180), within the 4,000–6,000-word guide for Regular Papers. Detailed one-at-a-time sweeps, illumination/dark characteristics, screening/setup tables, validation provenance, oxidation-budget derivation and a reproducibility appendix are provided as supplementary material (cited in text).

Highlights (Solar Energy submission — RbGeI3/TiO2/CuI SCAPS-1D study)

- Cu2O BSF lifts RbGeI3 stack 27.06% to 28.01%; +0.98 pp at full realism.
- Stacked loss window 20.78% (21.76% with BSF); D2 cross-check 28.16%.
- Oxidation budget: E^f>=0.46 eV holds 27.06%; ambient below 20.76%.
- SHAP ranks bulk defects, gap, electron mobility; R2 0.907 held out.
- SCAPS pipeline calibration-checked on CsPbI3 within 2% on both devices.

Ethics declaration

The author(s) declare(s) that the study does not involve humans nor animal subjects.

Click here to view linked References

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)

[1


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texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]

[2]

[3]
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FTO/T
iO$[]$/RbGeI$[]$/CuI/
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! Package pdftex.def Error: File `figs/fig2.png' not found: using draft
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See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
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l.64 ...phics[width=0.85\linewidth]{figs/fig2.png}
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If that doesn't work, type  X <return>  to quit.

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line 6
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)

[4]

[5]

[6]

[7]
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 []


Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/PCBM/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/
 []


Underfull \hbox (badness 10000) in paragraph at lines 160--160
\T1/cmr/m/n/8 TiO$[]$/RbGeI$[]$/NiO/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/PCBM/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
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\T1/cmr/m/n/8 TiO$[]$/RbGeI$[]$/CuI/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
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[]|\T1/cmr/m/n/8 FTO/PCBM/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/
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\T1/cmr/m/n/8 TiO$[]$/RbGeI$[]$/CBTS/
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Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/PCBM/
 []


Underfull \hbox (badness 10000) in paragraph at lines 160--160
[]|\T1/cmr/m/n/8 FTO/
 []


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\T1/cmr/m/n/8 TiO$[]$/RbGeI$[]$/Spiro/Au|
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[8]

[9]
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eened FTO/TiO$[]$/RbGeI$[]$/CuI/
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[10]

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! Package pdftex.def Error: File `figs/fig4.png' not found: using draft
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See the pdftex.def package documentation for explanation.
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If that doesn't work, type  X <return>  to quit.



[11]

[12]

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See the pdftex.def package documentation for explanation.
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g.

See the pdftex.def package documentation for explanation.
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Try typing  <return>  to proceed.
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[13]

[14]
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[15]

[16]

LaTeX Warning: File `figs/fig17.png' not found on input line 278.


! Package pdftex.def Error: File `figs/fig17.png' not found: using draft
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See the pdftex.def package documentation for explanation.
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g.

See the pdftex.def package documentation for explanation.
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 ...

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See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.295 ...ics[width=0.85\linewidth]{figs/fig21.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.
```

```
LaTeX Warning: File `figs/fig22.png' not found on input line 301.


! Package pdftex.def Error: File `figs/fig22.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.301 ...ics[width=0.85\linewidth]{figs/fig22.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig23.png' not found on input line 307.


! Package pdftex.def Error: File `figs/fig23.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.307 ...ics[width=0.85\linewidth]{figs/fig23.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[17]

[18]

[19]

[20]

[21]

[22]
Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []


Underfull \hbox (badness 10000) in paragraph at lines 340--340
```

\T1/cmr/m/n/7 TiO$[]$/RbGeI$[]$/NiO/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/TiO$[]$/CsGeI$[]$/CuI/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 ITO/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
\T1/cmr/m/n/7 C$[]$/RbGeI$[]$/CBTS/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []

Overfull \hbox (12.0532pt too wide) in paragraph at lines 340--340
\T1/cmr/m/n/7 TiO$[]$/RbGeI$[]$/PEDOT:PSS/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
\T1/cmr/m/n/7 TiO$[]$/MASnI$[]$/CBTS/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []

Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []

```
Underfull \hbox (badness 10000) in paragraph at lines 340--340
\T1/cmr/m/n/7 TiO$[]$/RbGeI$[]$/Spiro-
 []


Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []


Underfull \hbox (badness 10000) in paragraph at lines 340--340
\T1/cmr/m/n/7 TiO$[]$/RbGeI$[]$/CuI/
 []


Underfull \hbox (badness 10000) in paragraph at lines 340--340
[]|\T1/cmr/m/n/7 FTO/
 []


Underfull \hbox (badness 10000) in paragraph at lines 340--340
\T1/cmr/m/n/7 TiO$[]$/RbGeI$[]$/Cu$[]$O-
 []


Underfull \hbox (badness 10000) in paragraph at lines 359--359
[]|\T1/cmr/m/n/8 High ox-i-da-tion
 []


Underfull \hbox (badness 10000) in paragraph at lines 359--359
[]|\T1/cmr/m/n/8 High ox-i-da-tion
 []


LaTeX Warning: File `figs/fig24.png' not found on input line 366.


! Package pdftex.def Error: File `figs/fig24.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.366 ...ics[width=0.85\linewidth]{figs/fig24.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig25.png' not found on input line 372.
```

```
! Package pdftex.def Error: File `figs/fig25.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.372 ...ics[width=0.85\linewidth]{figs/fig25.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig26.png' not found on input line 378.


! Package pdftex.def Error: File `figs/fig26.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.378 ...ics[width=0.85\linewidth]{figs/fig26.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig27.png' not found on input line 384.


! Package pdftex.def Error: File `figs/fig27.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.384 ...ics[width=0.85\linewidth]{figs/fig27.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig28.png' not found on input line 390.


! Package pdftex.def Error: File `figs/fig28.png' not found: using draft
settin
```

```
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.390 ...ics[width=0.85\linewidth]{figs/fig28.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig29.png' not found on input line 396.


! Package pdftex.def Error: File `figs/fig29.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.396 ...ics[width=0.85\linewidth]{figs/fig29.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig30.png' not found on input line 402.


! Package pdftex.def Error: File `figs/fig30.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.402 ...ics[width=0.85\linewidth]{figs/fig30.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig31.png' not found on input line 408.


! Package pdftex.def Error: File `figs/fig31.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
```

```
Type  H <return>  for immediate help.
 ...

l.408 ...ics[width=0.85\linewidth]{figs/fig31.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[23]

[24]

[25]

[26]

[27]

[28]

[29]

[30]

[31]

[32]

LaTeX Warning: File `figs/fig32.png' not found on input line 417.


! Package pdftex.def Error: File `figs/fig32.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.417 ...ics[width=0.85\linewidth]{figs/fig32.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig23.png' not found on input line 426.


! Package pdftex.def Error: File `figs/fig23.png' not found: using draft
settin
g.
```

```
See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.426 ...ics[width=0.85\linewidth]{figs/fig23.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[33]

[34]

[35]

LaTeX Warning: File `figs/fig34.png' not found on input line 435.


! Package pdftex.def Error: File `figs/fig34.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.435 ...ics[width=0.85\linewidth]{figs/fig34.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[36]

LaTeX Warning: File `figs/fig35.png' not found on input line 444.


! Package pdftex.def Error: File `figs/fig35.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.444 ...ics[width=0.85\linewidth]{figs/fig35.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.
```

```
LaTeX Warning: File `figs/fig36.png' not found on input line 450.


! Package pdftex.def Error: File `figs/fig36.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.450 ...ics[width=0.85\linewidth]{figs/fig36.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[37]

[38]

[39]

LaTeX Warning: File `figs/fig37.png' not found on input line 476.


! Package pdftex.def Error: File `figs/fig37.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.476 ...raphics[width=\textwidth]{figs/fig37.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.


LaTeX Warning: File `figs/fig38.png' not found on input line 485.


! Package pdftex.def Error: File `figs/fig38.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.485 ...raphics[width=\textwidth]{figs/fig38.png}
```

```
Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[40]

[41]

[42]

LaTeX Warning: File `figs/fig39.png' not found on input line 494.


! Package pdftex.def Error: File `figs/fig39.png' not found: using draft
settin
g.

See the pdftex.def package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.494 ...raphics[width=\textwidth]{figs/fig39.png}

Try typing  <return>  to proceed.
If that doesn't work, type  X <return>  to quit.



[43]

[44]

[45]

[46]

[47]

[48]

[49]

[50]

[51]

[52]
Underfull \hbox (badness 1112) in paragraph at lines 573--574
\T1/cmr/m/n/10 sim-u-lated PCE,'' J. Mater. Sci.: Mater. Eng. (2026).
doi:$\T1/
cmtt/m/n/10 10 . 1186 /
 []
```

```
[53]
Underfull \hbox (badness 1092) in paragraph at lines 581--582
[]\T1/cmr/m/n/10 W. J. Jang, P. J. Park, W.-J. Ong and S. Y. Kim, ``Sta-
ble and

 []



[54]

[55] (./manuscript.aux)
 ***********
LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-05-27>
 ***********
 )
Here is how much of TeX's memory you used:
 5818 strings out of 473583
 90800 string characters out of 5732343
 1947908 words of memory out of 5000000
 28632 multiletter control sequences out of 15000+600000
 574466 words of font info for 80 fonts, out of 8000000 for 9000
 1141 hyphenation exceptions out of 8191
 72i,13n,79p,2470b,507s stack positions out of
10000i,1000n,20000p,200000b,200000s
<c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmex10.pfb><c:/tex
live/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi10.pfb><c:/texlive/2024
/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi6.pfb><c:/texlive/2024/texmf-dis
t/fonts/type1/public/amsfonts/cm/cmmi7.pfb><c:/texlive/2024/texmf-
dist/fonts/ty
pe1/public/amsfonts/cm/cmmi8.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public
/amsfonts/cm/cmr10.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/
cm/cmr5.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmr6.pfb
><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmr7.pfb><c:/texliv
e/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmr8.pfb><c:/texlive/2024/texm
f-dist/fonts/type1/public/amsfonts/cm/cmsy10.pfb><c:/texlive/2024/texmf-
dist/fo
nts/type1/public/amsfonts/cm/cmsy6.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/
public/amsfonts/cm/cmsy7.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/ams
fonts/cm/cmsy8.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/symb
```

```
ols/msam10.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfbx1000
.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfrm0600.pfb><c:/t
exlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfrm0700.pfb><c:/texlive/202
4/texmf-dist/fonts/type1/public/cm-
super/sfrm0800.pfb><c:/texlive/2024/texmf-di
st/fonts/type1/public/cm-super/sfrm0900.pfb><c:/texlive/2024/texmf-
dist/fonts/t
ype1/public/cm-super/sfrm1000.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/publi
c/cm-super/sfrm1440.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/cm-super
/sfti0600.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfti0800.
pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfti1000.pfb><c:/te
xlive/2024/texmf-dist/fonts/type1/public/cm-super/sftt1000.pfb>
Output written on manuscript.pdf (55 pages, 519265 bytes).
PDF statistics:
 315 PDF objects out of 1000 (max. 8388607)
 203 compressed objects within 3 object streams
 0 named destinations out of 1000 (max. 500000)
 1 words of extra memory for PDF output out of 10000 (max. 10000000)
```

Table Click here to access/download;Table;tables.tex

```
This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2024)
(preloaded format=pdflatex 2024.8.2)  12 OCT 2026 00:42
entering extended mode
 restricted \write18 enabled.
 %&-line parsing enabled.
**tables.tex
(./tables.tex
LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-05-27>
(c:/texlive/2024/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/02/08 v1.4n Standard LaTeX document class
(c:/texlive/2024/texmf-dist/tex/latex/base/size12.clo
File: size12.clo 2024/02/08 v1.4n Standard LaTeX file (size option)
)
\c@part=\count194
\c@section=\count195
\c@subsection=\count196
\c@subsubsection=\count197
\c@paragraph=\count198
\c@subparagraph=\count199
\c@figure=\count266
\c@table=\count267
\abovecaptionskip=\skip49
\belowcaptionskip=\skip50
\bibindent=\dimen141
) (c:/texlive/2024/texmf-dist/tex/latex/base/inputenc.sty
Package: inputenc 2024/02/08 v1.3d Input encoding file
\inpenc@prehook=\toks17
\inpenc@posthook=\toks18
) (c:/texlive/2024/texmf-dist/tex/latex/base/fontenc.sty
Package: fontenc 2021/04/29 v2.0v Standard LaTeX package
) (c:/texlive/2024/texmf-dist/tex/latex/geometry/geometry.sty
Package: geometry 2020/01/02 v5.9 Page Geometry
(c:/texlive/2024/texmf-dist/tex/latex/graphics/keyval.sty
Package: keyval 2022/05/29 v1.15 key=value parser (DPC)
\KV@toks@=\toks19
) (c:/texlive/2024/texmf-dist/tex/generic/iftex/ifvtex.sty
Package: ifvtex 2019/10/25 v1.7 ifvtex legacy package. Use iftex instead.
(c:/texlive/2024/texmf-dist/tex/generic/iftex/iftex.sty
Package: iftex 2022/02/03 v1.0f TeX engine tests
))
\Gm@cnth=\count268
\Gm@cntv=\count269
\c@Gm@tempcnt=\count270
\Gm@bindingoffset=\dimen142
\Gm@wd@mp=\dimen143
\Gm@odd@mp=\dimen144
\Gm@even@mp=\dimen145
\Gm@layoutwidth=\dimen146
\Gm@layoutheight=\dimen147
\Gm@layouthoffset=\dimen148
\Gm@layoutvoffset=\dimen149
\Gm@dimlist=\toks20
) (c:/texlive/2024/texmf-dist/tex/latex/tools/tabularx.sty
```

```
Package: tabularx 2023/12/11 v2.12a `tabularx' package (DPC)
(c:/texlive/2024/texmf-dist/tex/latex/tools/array.sty
Package: array 2024/06/14 v2.6d Tabular extension package (FMi)
\col@sep=\dimen150
\ar@mcellbox=\box52
\extrarowheight=\dimen151
\NC@list=\toks21
\extratabsurround=\skip51
\backup@length=\skip52
\ar@cellbox=\box53
)
\TX@col@width=\dimen152
\TX@old@table=\dimen153
\TX@old@col=\dimen154
\TX@target=\dimen155
\TX@delta=\dimen156
\TX@cols=\count271
\TX@ftn=\toks22
) (c:/texlive/2024/texmf-dist/tex/latex/booktabs/booktabs.sty
Package: booktabs 2020/01/12 v1.61803398 Publication quality tables
\heavyrulewidth=\dimen157
\lightrulewidth=\dimen158
\cmidrulewidth=\dimen159
\belowrulesep=\dimen160
\belowbottomsep=\dimen161
\aboverulesep=\dimen162
\abovetopsep=\dimen163
\cmidrulesep=\dimen164
\cmidrulekern=\dimen165
\defaultaddspace=\dimen166
\@cmidla=\count272
\@cmidlb=\count273
\@aboverulesep=\dimen167
\@belowrulesep=\dimen168
\@thisruleclass=\count274
\@lastruleclass=\count275
\@thisrulewidth=\dimen169
) (c:/texlive/2024/texmf-dist/tex/latex/graphics/graphicx.sty
Package: graphicx 2021/09/16 v1.2d Enhanced LaTeX Graphics (DPC,SPQR)
(c:/texlive/2024/texmf-dist/tex/latex/graphics/graphics.sty
Package: graphics 2024/05/23 v1.4g Standard LaTeX Graphics (DPC,SPQR)
(c:/texlive/2024/texmf-dist/tex/latex/graphics/trig.sty
Package: trig 2023/12/02 v1.11 sin cos tan (DPC)
) (c:/texlive/2024/texmf-dist/tex/latex/graphics-cfg/graphics.cfg
File: graphics.cfg 2016/06/04 v1.11 sample graphics configuration
)
Package graphics Info: Driver file: pdftex.def on input line 106.
(c:/texlive/2024/texmf-dist/tex/latex/graphics-def/pdftex.def
File: pdftex.def 2024/04/13 v1.2c Graphics/color driver for pdftex
))
\Gin@req@height=\dimen170
\Gin@req@width=\dimen171
) (c:/texlive/2024/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def
```

```
File: l3backend-pdftex.def 2024-05-08 L3 backend support: PDF output
(pdfTeX)
\l__color_backend_stack_int=\count276
\l__pdf_internal_box=\box54
) (./tables.aux)
\openout1 = `tables.aux'.

LaTeX Font Info:    Checking defaults for OML/cmm/m/it on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for OMS/cmsy/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for OT1/cmr/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for T1/cmr/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for TS1/cmr/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for OMX/cmex/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
LaTeX Font Info:    Checking defaults for U/cmr/m/n on input line 7.
LaTeX Font Info:    ... okay on input line 7.
*geometry* driver: auto-detecting
*geometry* detected driver: pdftex
*geometry* verbose mode - [ preamble ] result:
* driver: pdftex
* paper: <default>
* layout: <same size as paper>
* layoutoffset:(h,v)=(0.0pt,0.0pt)
* modes:
* h-part:(L,W,R)=(72.26999pt, 469.75502pt, 72.26999pt)
* v-part:(T,H,B)=(72.26999pt, 650.43001pt, 72.26999pt)
* \paperwidth=614.295pt
* \paperheight=794.96999pt
* \textwidth=469.75502pt
* \textheight=650.43001pt
* \oddsidemargin=0.0pt
* \evensidemargin=0.0pt
* \topmargin=-37.0pt
* \headheight=12.0pt
* \headsep=25.0pt
* \topskip=12.0pt
* \footskip=30.0pt
* \marginparwidth=44.0pt
* \marginparsep=10.0pt
* \columnsep=10.0pt
* \skip\footins=10.8pt plus 4.0pt minus 2.0pt
* \hoffset=0.0pt
* \voffset=0.0pt
* \mag=1000
* \@twocolumnfalse
* \@twosidefalse
* \@mparswitchfalse
* \@reversemarginfalse
* (1in=72.27pt=25.4mm, 1cm=28.453pt)
```

```
(c:/texlive/2024/texmf-dist/tex/context/base/mkii/supp-pdf.mkii
[Loading MPS to PDF converter (version 2006.09.02).]
\scratchcounter=\count277
\scratchdimen=\dimen172
\scratchbox=\box55
\nofMPsegments=\count278
\nofMParguments=\count279
\everyMPshowfont=\toks23
\MPscratchCnt=\count280
\MPscratchDim=\dimen173
\MPnumerator=\count281
\makeMPintoPDFobject=\count282
\everyMPtoPDFconversion=\toks24
) (c:/texlive/2024/texmf-dist/tex/latex/epstopdf-pkg/epstopdf-base.sty
Package: epstopdf-base 2020-01-24 v2.11 Base part for package epstopdf
Package epstopdf-base Info: Redefining graphics rule for `.eps' on input
line 4
85.
(c:/texlive/2024/texmf-dist/tex/latex/latexconfig/epstopdf-sys.cfg
File: epstopdf-sys.cfg 2010/07/13 v1.3 Configuration of (r)epstopdf for
TeX Liv
e
))
Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/PCBM/RbGeI$[]$/NiO/
 []


Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/PCBM/RbGeI$[]$/CuI/
 []


Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/PCBM/
 []


Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/TiO$[]$/RbGeI$[]$/CBTS/
 []


Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/PCBM/
 []


Underfull \hbox (badness 10000) in paragraph at lines 46--46
[]|\T1/cmr/m/n/10 FTO/
 []

LaTeX Font Info:    External font `cmex10' loaded for size
```

```
(Font)              <12> on input line 94.
! Undefined control sequence.
<argument> ... 8; optics row reports PCE$_{1\text
                                                    {-sun}}$). PCEs are
unroun...
l.111 ...CAPS outputs (displayed inputs rounded).}

The control sequence at the end of the top line
of your error message was never \def'ed. If you have
misspelled it (e.g., `\hobx'), type `I' and the correct
spelling (e.g., `I\hbox'). Otherwise just continue,
and I'll forget about whatever was undefined.

! Undefined control sequence.
<argument> ... 8; optics row reports PCE$_{1\text
                                                    {-sun}}$). PCEs are
unroun...
l.111 ...CAPS outputs (displayed inputs rounded).}

The control sequence at the end of the top line
of your error message was never \def'ed. If you have
misspelled it (e.g., `\hobx'), type `I' and the correct
spelling (e.g., `I\hbox'). Otherwise just continue,
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                                                    {-sun}}$). PCEs are
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misspelled it (e.g., `\hobx'), type `I' and the correct
spelling (e.g., `I\hbox'). Otherwise just continue,
and I'll forget about whatever was undefined.


LaTeX Warning: Citation `r14' on page 1 undefined on input line 132.


LaTeX Warning: Citation `r24' on page 1 undefined on input line 132.


LaTeX Warning: Citation `r22' on page 1 undefined on input line 132.


LaTeX Warning: Citation `r39' on page 1 undefined on input line 132.


LaTeX Warning: Citation `r41' on page 1 undefined on input line 132.


LaTeX Warning: Citation `r42' on page 1 undefined on input line 132.
```

```
LaTeX Warning: Citation `r40' on page 1 undefined on input line 132.


Underfull \hbox (badness 10000) in paragraph at lines 132--132
[]|\T1/cmr/m/n/8 FTO/TiO$[]$/RbGeI$[]$/Spiro-OMeTAD/
 []


LaTeX Warning: Citation `r23' on page 1 undefined on input line 132.


Underfull \hbox (badness 10000) in paragraph at lines 132--132
[]|\T1/cmr/m/n/8 FTO/TiO$[]$/RbGeI$[]$/CuI/Au (this
 []


Underfull \hbox (badness 10000) in paragraph at lines 132--132
[]|\T1/cmr/m/n/8 FTO/TiO$[]$/RbGeI$[]$/Cu$[]$O-BSF/CuI/
 []


LaTeX Warning: Citation `rWang' on page 1 undefined on input line 170.


LaTeX Warning: Citation `rWang' on page 1 undefined on input line 170.



[1

{c:/texlive/2024/texmf-
var/fonts/map/pdftex/updmap/pdftex.map}{c:/texlive/2024/
texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]

[2]

[3] (./tables.aux)
 ***********
LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-05-27>
 ***********

LaTeX Warning: There were undefined references.

 )
Here is how much of TeX's memory you used:
 2311 strings out of 473583
 35043 string characters out of 5732343
 1962908 words of memory out of 5000000
 25208 multiletter control sequences out of 15000+600000
 567878 words of font info for 56 fonts, out of 8000000 for 9000
 1141 hyphenation exceptions out of 8191
```

```
 57i,13n,65p,430b,360s stack positions out of
10000i,1000n,20000p,200000b,200000s
<c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi10.pfb><c:/tex
live/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi12.pfb><c:/texlive/2024
/texmf-
dist/fonts/type1/public/amsfonts/cm/cmmi7.pfb><c:/texlive/2024/texmf-dis
t/fonts/type1/public/amsfonts/cm/cmmi8.pfb><c:/texlive/2024/texmf-
dist/fonts/ty
pe1/public/amsfonts/cm/cmr6.pfb><c:/texlive/2024/texmf-
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/cmr8.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmsy10.pfb
><c:/texlive/2024/texmf-
dist/fonts/type1/public/amsfonts/cm/cmsy7.pfb><c:/texli
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dist/fonts/type1/public/amsfonts/cm/cmsy8.pfb><c:/texlive/2024/te
xmf-dist/fonts/type1/public/cm-super/sfbx0800.pfb><c:/texlive/2024/texmf-
dist/f
onts/type1/public/cm-super/sfbx1200.pfb><c:/texlive/2024/texmf-
dist/fonts/type1
/public/cm-super/sfbx1728.pfb><c:/texlive/2024/texmf-
dist/fonts/type1/public/cm
-super/sfrm0800.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfr
m1000.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-
super/sfrm1095.pfb>
<c:/texlive/2024/texmf-dist/fonts/type1/public/cm-super/sfrm1200.pfb>
Output written on tables.pdf (3 pages, 227106 bytes).
PDF statistics:
 100 PDF objects out of 1000 (max. 8388607)
 60 compressed objects within 1 object stream
 0 named destinations out of 1000 (max. 500000)
 1 words of extra memory for PDF output out of 10000 (max. 10000000)
```

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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